

# US Patent & Trademark Office

## Patent Public Search | Text View

---

United States Patent

12405180

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Gardner; Eric M. et al.

---

### System and method for strain and acceleration based analytics in an independent cart system

---

#### Abstract

A system and method of monitoring forces exerted at multiple locations on a mover includes multiple sensors, where each sensor is mounted at one of the locations. Each sensor detects an operating condition of the mover at the location on the mover at which it is mounted. The sensors may include accelerometers, strain gauges, or a combination thereof. Each strain gauge is mounted proximate to an area of interest on the mover. Each strain gauge generates a feedback signal corresponding to a deformation of the material measured at the location of the sensor. From the measured deformation of material, a force acting on the mover at the location of the sensor may be determined. The forces exerted at the different locations on the mover may be monitored in real time to determine bearing performance or monitored over a duration of time to observe changes in bearing performance over that duration.

---

**Inventors:** Gardner; Eric M. (Milwaukee, WI), Ramanarayanan; Shankernarayan (Eden Prairie, MN), Rajavel; Tamil K. (Carver, MN)

**Applicant:** Rockwell Automation Technologies, Inc. (Mayfield Heights, OH)

**Family ID:** 83360882

**Assignee:** Rockwell Automation Technologies, Inc. (Mayfield Heights, OH)

**Appl. No.:** 18/635827

**Filed:** April 15, 2024

#### Prior Publication Data

**Document Identifier**

**Publication Date**

US 20240255368 A1

Aug. 01, 2024

#### Related U.S. Application Data

## Publication Classification

**Int. Cl.:** H04B17/309 (20150101); G01L5/00 (20060101); G01L5/04 (20060101)

**U.S. Cl.:**

**CPC** G01L5/0019 (20130101); G01L5/0028 (20130101); G01L5/042 (20130101);

## Field of Classification Search

**CPC:** G01L (5/0019); G01L (5/0028); G01L (5/042); B60L (15/005); B65G (43/00)

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name        | U.S. Cl. | CPC         |
|--------------|-------------|----------------------|----------|-------------|
| 5241912      | 12/1992     | Oshima et al.        | N/A      | N/A         |
| 7486182      | 12/2008     | Hahn et al.          | N/A      | N/A         |
| 7890194      | 12/2010     | Pannese              | N/A      | N/A         |
| 9673685      | 12/2016     | Cooper et al.        | N/A      | N/A         |
| 10135705     | 12/2017     | Asenjo et al.        | N/A      | N/A         |
| 10432117     | 12/2018     | Huang et al.         | N/A      | N/A         |
| 10488282     | 12/2018     | Guru et al.          | N/A      | N/A         |
| 10586254     | 12/2019     | Singhal              | N/A      | N/A         |
| 10678233     | 12/2019     | Cella et al.         | N/A      | N/A         |
| 10712738     | 12/2019     | Cella et al.         | N/A      | N/A         |
| 10717643     | 12/2019     | Yee et al.           | N/A      | N/A         |
| 10866584     | 12/2019     | Cella et al.         | N/A      | N/A         |
| 2003/0121333 | 12/2002     | Smith                | 73/796   | G01L 5/0028 |
| 2014/0025256 | 12/2013     | Armitage et al.      | N/A      | N/A         |
| 2014/0067289 | 12/2013     | Baldwin              | N/A      | N/A         |
| 2014/0074433 | 12/2013     | Crepet               | N/A      | N/A         |
| 2018/0321135 | 12/2017     | Zimmerman            | N/A      | G01N 3/08   |
| 2019/0033846 | 12/2018     | Cella et al.         | N/A      | N/A         |
| 2019/0061558 | 12/2018     | Craver               | N/A      | N/A         |
| 2019/0250069 | 12/2018     | Samadani et al.      | N/A      | N/A         |
| 2019/0298084 | 12/2018     | Billings et al.      | N/A      | N/A         |
| 2020/0132149 | 12/2019     | Chandrasekara et al. | N/A      | N/A         |

### FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 110595775  | 12/2018          | CN      | N/A |
| 2592515    | 12/2012          | EP      | N/A |
| 2801938    | 12/2013          | EP      | N/A |
| 2841781    | 12/2017          | EP      | N/A |
| 3299317    | 12/2017          | EP      | N/A |
| 3628605    | 12/2019          | EP      | N/A |

|            |         |    |     |
|------------|---------|----|-----|
| 2017196821 | 12/2016 | WO | N/A |
| 2020036818 | 12/2019 | WO | N/A |
| 2020097562 | 12/2019 | WO | N/A |

## OTHER PUBLICATIONS

European Patent Office, Communication pursuant to Article 94(3) EPC, EP Pat. Application No. 22 196 127.9, Feb. 5, 2025, 6 pages. cited by applicant

Extended European Search Report for Application No. 22196127.9, dated Feb. 17, 2023 (8 pages). cited by applicant

---

*Primary Examiner:* Islam; Muhammad S

*Attorney, Agent or Firm:* Boyle Fredrickson, S.C.

---

## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to U.S. application Ser. No. 17/482,746, filed Sep. 23, 2021, the entire contents of which is incorporated herein by reference.

### BACKGROUND INFORMATION

(2) The subject matter disclosed herein relates to a system and method for monitoring vehicle health for an independent cart system. More specifically, multiple sensors, such as strain gauges and accelerometers are mounted to different locations on a vehicle to provide performance and health of the vehicle or subsystems on the vehicle at the different locations.

(3) Motion control systems utilizing independent carts and linear motors can be used in a wide variety of processes (e.g. packaging, manufacturing, and machining) and can provide an advantage over conventional conveyor belt systems with enhanced flexibility, extremely high-speed movement, and mechanical simplicity. The motion control system includes a set of independently controlled vehicles or carts, also referred to herein as “movers”, each supported on a track for motion along the track. The track is made up of a number of track segments, and a linear drive system controls operation of the movers, causing the movers to travel along the track. Sensors may be spaced at fixed positions along the track and/or on the movers to provide information about the position and speed of the movers.

(4) Numerous differences between different movers or differences in a single mover over time may impact how a mover travels along the track. Variations between movers due, for example, to manufacturing tolerances may result in differences in physical engagement of the mover with the track. The variations in physical engagement may result in greater pressure and/or friction being experienced by one of the bearings on the mover than by other bearings. The bearing experiencing the greatest pressure and/or friction may wear more quickly than the other bearings. Similarly, variations in orientation of the track will result in forces due to gravity being exerted on the movers differently. Variations in loading on each mover as it travels along the track will cause varying forces to be exerted on each mover. All of the variations in manufacture, orientation, loading, and the like impact external forces experienced by a mover and may similarly result in greater pressure and/or friction being experienced by one of the bearings on the mover than by other bearings.

(5) Thus, it would be desirable to monitor forces exerted at multiple locations on a mover.

(6) Further, wear in bearings or rollers over time may increase variations between different movers or change orientation of a single mover over time. Wear on the track, such as a dent on a rail, or varying transition distances between track segments may further impact how a mover travels along

the track.

(7) Thus, it would be desirable to have real-time feedback corresponding to bearing performance, and to monitor changes in bearing performance over time.

#### BRIEF DESCRIPTION

(8) According to one embodiment of the invention, a system for monitoring status of a mover in an independent cart system includes multiple sensors, a control circuit, and a transmitter. The independent cart system includes multiple movers configured to travel along a track. The sensors are mounted on the movers, and each sensor is configured to generate at least one feedback signal corresponding to an operating condition of the mover. The control circuit is mounted on the mover. The control circuit is configured to receive the at least one feedback signal from each of the sensors and to generate a data packet including a value corresponding to the operating condition monitored from each of the sensors. The transmitter is mounted on the mover. The transmitter is configured to receive the data packet from the control circuit and to transmit the data packet to a receiver located external from the mover.

(9) According to another embodiment of the invention, a system for monitoring status of multiple bearings on a mover in an independent cart system includes multiple sensors, a control circuit, and a transmitter. The sensors are mounted on the mover, and each sensor is mounted proximate one of the bearings on the mover. Each sensor is configured to generate at least one feedback signal corresponding to an operating condition of a corresponding bearing by which each of the sensors is mounted. The control circuit is mounted on the mover. The control circuit is configured to receive the at least one feedback signal from each of the sensors and to generate a data packet including a value for the at least one feedback signal corresponding to the operating condition of the corresponding bearing monitored from each of the sensors. The transmitter is mounted on the mover. The transmitter is configured to receive the data packet from the control circuit and to transmit the data packet to a receiver located external from the mover.

(10) According to still another embodiment of the invention, a method for monitoring status of a mover in an independent cart system includes generating at least one feedback signal from each of multiple sensors mounted on the mover. The at least one feedback signal corresponds to an operating condition of the mover and is received from each of the sensors at a control circuit mounted on the mover. The control circuit generates a data packet including at least one value corresponding to the operating condition monitored by each of the sensors. The data packet is transmitted from a transmitter mounted on the mover to a receiver located external from the mover.

(11) These and other advantages and features of the invention will become apparent to those skilled in the art from the detailed description and the accompanying drawings. It should be understood, however, that the detailed description and accompanying drawings, while indicating preferred embodiments of the present invention, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the present invention without departing from the spirit thereof, and the invention includes all such modifications.

---

## Description

#### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various exemplary embodiments of the subject matter disclosed herein are illustrated in the accompanying drawings in which like reference numerals represent like parts throughout, and in which:

(2) FIG. 1 is a perspective view of an exemplary linear cart system incorporating multiple movers travelling along a closed curvilinear track according to one embodiment of the present invention;

(3) FIG. 2 is a perspective view of an exemplary linear cart system incorporating multiple movers travelling along a closed curvilinear track according to another embodiment of the present

invention;

(4) FIG. 3 is a partial side elevation of the linear cart system of FIG. 2;

(5) FIG. 4 is a perspective view of a mover from the transport system of FIG. 1;

(6) FIG. 5 is a partial sectional view of the transport system of FIG. 1;

(7) FIG. 6 is a side elevational view of a mover from the transport system of FIG. 2;

(8) FIG. 7 is a partial sectional view of the transport system of FIG. 2;

(9) FIG. 8 is a partial side elevation view of one segment of one embodiment of the linear cart system of FIG. 1 illustrating activation coils distributed along one surface of the track segment;

(10) FIG. 9 is an exemplary control system for a linear cart system incorporating one embodiment of the present invention;

(11) FIG. 10 is a block diagram representation of the control system of FIG. 9;

(12) FIG. 11 is a perspective view of a mover from the transport system of FIG. 2, illustrating multiple sensors and a control circuit mounted to the mover according to one embodiment of the invention;

(13) FIG. 12 is a schematic representation of an exemplary strain gauge utilized as one of the sensors of FIG. 11;

(14) FIG. 13 is a block diagram representation of an exemplary control circuit of FIG. 11;

(15) FIG. 14 is a front elevation view of an exemplary track mounted with the major axis of the track positioned in a vertical orientation;

(16) FIG. 15 is a front elevation view of an exemplary track mounted in a horizontal orientation; and

(17) FIG. 16 is a front elevation view of an exemplary track mounted with the minor axis of the track positioned in a vertical orientation.

(18) In describing the various embodiments of the invention which are illustrated in the drawings, specific terminology will be resorted to for the sake of clarity. However, it is not intended that the invention be limited to the specific terms so selected and it is understood that each specific term includes all technical equivalents which operate in a similar manner to accomplish a similar purpose. For example, the word "connected," "attached," or terms similar thereto are often used. They are not limited to direct connection but include connection through other elements where such connection is recognized as being equivalent by those skilled in the art.

#### DETAILED DESCRIPTION

(19) The various features and advantageous details of the subject matter disclosed herein are explained more fully with reference to the non-limiting embodiments described in detail in the following description. The subject matter disclosed herein describes a system and method of monitoring forces exerted at multiple locations on a mover. Multiple sensors are mounted on each mover at multiple locations. Each sensor detects an operating condition of the mover at the location on the mover at which it is mounted. According to one embodiment of the invention, at least one accelerometer and at least one strain gauge are mounted to the mover. According to one aspect of the invention, multiple strain gauges may be mounted to the mover, where each strain gauge is mounted proximate to one of the bearings of the mover. Each strain gauge generates a feedback signal corresponding to a deformation of the material measured at the location of the sensor. From the measured deformation of material, a force acting on the mover at the location of the sensor may be determined. The strain gauges, therefore, can be used to monitor forces exerted at multiple locations on the mover. According to one aspect of the invention, the forces exerted at multiple locations on the mover may be monitored in real time to determine bearing performance. The forces may also be monitored over a duration of time to observe changes in bearing performance over that duration.

(20) Each sensor is configured to generate a feedback signal corresponding to the measured strain. It is contemplated that the feedback signals, additional processed signals, or a combination thereof may be transmitted from each mover to a remote device. Each mover travels along the track of the

independent cart system. The remote device is preferably a stationary device mounted external from the track and may, for example, provide a visual interface for a user on which the real-time forces experienced at each location on the mover are displayed. On a smaller track, a communication bus may be mounted around the track and each cart may include a brush, configured to slide along the communication bus and to establish a “wired” communication path between the cart and the remote device. However, with branches and/or an increased size in the track, a wired communication path becomes impractical and a wireless communication path is preferred. Each mover may include a wireless communication device in communication with the sensors and with the remote device to transfer data directly from the sensors to the remote device. Optionally, intermediate communication nodes may be established periodically along the track allowing a wireless communication device on each mover to communicate to one of the nodes. Each node may, in turn, be connected via a wired connection, a wireless connection, or a combination thereof to the remote device. Each node may serve as a gateway to transfer data between the carts and the remote device.

(21) In one embodiment of the invention, it is contemplated that power for each of the sensors and for the communication device is provided by a battery mounted on the cart. According to another embodiment of the invention, it is contemplated that power for each of the sensors and for the communication device is provided via a wireless power transfer system. The wireless power transfer system may use, for example, inductive or optical coupling between a power source mounted on or adjacent to the track, where the power source is configured to emit energy to a pickup device mounted on the cart. The pickup device is configured to receive the emitted energy when it is range of the power source. Multiple power sources may be stationed around the track or, optionally, a power rail may be mounted continuously along the track and a power pick-up may be inductively coupled to the power rail to continuously receive power on the cart from the wireless power transfer system. The cart may have an energy storage device in which energy transferred to the cart is stored until a sensor, control circuit, or communication device requires energy for activation.

(22) Turning initially to FIGS. **1-3**, two embodiments of an exemplary transport system for moving articles or products are illustrated. The transport system includes a track **10** made up of multiple segments **12**, **14**. According to the illustrated embodiments, the segments define a generally closed loop supporting a set of movers **100** movable along the track **10**. The illustrated tracks **10** each include four straight segments **12** with two straight segments **12** located along each side of the track and spaced apart from the other pair. The tracks **10** also include four curved segments **14** where a pair of curved segments **14** is located at each end of the track **10** to connect the pairs of straight segments **12**. The four straight segments **12** and the four curved segments **14** form a generally oval track and define a closed path over which each of the movers **100** may travel. It is understood that track segments of various sizes, lengths, and shapes may be connected together to form a track **10** without deviating from the scope of the invention.

(23) In FIG. **1**, the track **10** is oriented in a horizontal plane and supported above the ground by a base **15** extending vertically downward from the track **10**. The base **15** includes a pair of generally planar support plates **17**, located on opposite sides of the track **10**, with mounting feet **19** on each support plate **17** to secure the track **10** to a surface. In FIG. **2**, the track **10** is shown without a base. It is contemplated that the tracks **10** may be installed in different orientations, such as sloped or vertical, and include different shaped segments including, but not limited to, straight segments, inward bends, outward bends, up slopes, down slopes and various combinations thereof. For convenience, the horizontal orientation of the track **10** shown in FIG. **1** will be discussed herein. Terms such as upper, lower, inner, and outer will be used with respect to the illustrated track orientation. These terms are relational with respect to the illustrated track and are not intended to be limiting. The movers **100** will travel along the track and take various orientations according to the configuration of the track **10** and the relationships discussed herein may vary accordingly.

(24) Each track segment **12**, **14** includes a number of independently attached rails **20** on which each mover **100** runs. According to the illustrated embodiments, rails **20** extend generally along the outer periphery of the track **10**. A first rail **20** extends along an upper surface **11** of each segment and a second rail **20** extends along a lower surface **13** of each segment. It is contemplated that each rail **20** may be a singular member, which may be molded, extruded, or machined as a single rail member, or each rail **20** may be assembled and formed from multiple members. It is also contemplated that the cross section of the rails **20** may be circular, square, rectangular, or any other desired cross-sectional shape without deviating from the scope of the invention. The rails **20** generally conform to the curvature of the track **10** thus extending in a straight path along the straight track segments **12** and in a curved path along the curved track segments **14**. The rails **20** may be thin with respect to the dimensions of the track **10** and span only a partial width of the surface of the track **10** on which it is attached.

(25) With reference also to FIG. 5, a first embodiment of the rail **20** includes a base portion **22** mounted to the track segment and a track portion **24** along which the mover **100** runs. Each mover **100** includes complementary rollers **110** to engage the track portion **24** of the rail **20** for movement along the track **10**. Each side of the track portion **24** is wedge-shaped and each roller **110** includes a complementary groove configured to receive the wedge-shaped side of the track portion.

(26) With reference also to FIG. 7, a second embodiment of the rail **20** includes two track portions **26**, **28**, where a first track portion **26** and a second track portion **28** each have a generally rectangular sectional area. The first track portion **26** of the upper rail **20** is positioned on the top surface **11** of the track and first and second rollers **110** engage each side of the first track portion **26** of the upper rail. The second track portion **28** of the upper rail **20** protrudes from the side of the track segment orthogonally to the orientation of the first track portion **26**. A third roller **110** engages one surface of the second track portion **28** of the upper rail. The first track portion **26** of a lower rail **20** similarly has a generally rectangular sectional area and is oriented on the lower surface **13** of the track and fourth and fifth rollers **110** engage each side of the first track portion **26** of the lower rail. The second track portion **28** of the lower rail **20** protrudes from the side of the track segment orthogonally to the orientation of the first track portion **26**, and a sixth roller **110** engages one surface of the second track portion **28** of the lower rail.

(27) One or more movers **100** are mounted to and movable along the rails **20** on the track **10**. With reference again to FIGS. 4 and 5, a first embodiment of an exemplary mover **100** is illustrated. Each mover **100** includes a side member **102**, a top member **104**, and a bottom member **106**. The side member **102** extends for a height at least spanning a distance between the rail **20** on the top surface **11** of the track **10** and the rail **20** on the bottom surface **13** of the track **10** and is oriented generally parallel to a side surface **21** when mounted to the track **10**. The top member **104** extends generally orthogonal to the side member **102** at a top end of the side member **102** and extends across the rail **20** on the top surface **11** of the track **10**. The top member **104** includes a first segment **103**, extending orthogonally from the side member **102** for the width of the rail **20**, which is generally the same width as the side member **102**. A set of rollers **110** are mounted on the lower side of the first segment **103** and are configured to engage the track portion **24** of the rail **20** mounted to the upper surface **11** of the track segment. According to the illustrated embodiment two pairs of rollers **110** are mounted to the lower side of the first segment **103** with a first pair located along a first edge of the track portion **24** of the rail and a second pair located along a second edge of the track portion **24** of the rail **20**. The first and second edges and, therefore, the first and second pairs of rollers **110** are on opposite sides of the rail **20** and positively retain the mover **100** to the rail **20**. The bottom member **106** extends generally orthogonal to the side member **102** at a bottom end of the side member **102** and extends for a distance sufficient to receive a third pair of rollers **110** along the bottom of the mover **100**. The third pair of rollers **110** engage an outer edge of the track portion **24** of the rail **20** mounted to the lower surface **13** of the track segment. Thus, the mover **100** rides along the rails **20** on the rollers **110** mounted to both the top member **104** and the

bottom member **106** of each mover **100**. The top member **104** also includes a second segment **120** which protrudes from the first segment **103** an additional distance beyond the rail **20** and is configured to hold a position magnet **130**. According to the illustrated embodiment, the second segment **120** of the top member **104** includes a first portion **122** extending generally parallel to the rail **20** and tapering to a smaller width than the first segment **103** of the top member **104**. The second segment **120** also includes a second portion **124** extending downward from and generally orthogonal to the first portion **122**. The second portion **124** extends downward a distance less than the distance to the upper surface **11** of the track segment but of sufficient distance to have the position magnet **130** mounted thereto. According to the illustrated embodiment, a position magnet **130** is mounted within a recess **126** on the second portion **124** and is configured to align with a sensor **150** mounted within the top surface **11** of the track segment.

(28) With reference again to FIG. 7, a second embodiment of an exemplary mover **100** is illustrated. Each mover **100** includes a side member **102**, a top member **104**, and a bottom member **106**. The side member **102** extends for a height at least spanning a distance between the rail **20** on the top surface **11** of the track **10** and the rail **20** on the bottom surface **13** of the track **10** and is oriented generally parallel to a side surface **21** when mounted to the track **10**. The top member **104** extends generally orthogonal to the side member **102** at a top end of the side member **102** and extends across the rail **20** on the top surface **11** of the track **10**. A first set of rollers **110** are mounted on the lower side of the top member **104** and are configured to engage either side of the first track portion **26** of the rail **20** mounted to the upper surface **11** of the track segment. According to the illustrated embodiment two pairs of rollers **110** are mounted to the lower side of the top member **104** with a first pair located along a first side of the first track portion **26** and a second pair located along a second side of the first track portion **26** of the upper rail **20**. A third pair of rollers **110** are mounted on the side member **102** and extend below the second track portion **28** of the upper rail. The bottom member **106** extends generally orthogonal to the side member **102** at a bottom end of the side member **102** and extends for a distance sufficient to receive a fourth and fifth pair of rollers **110** along the bottom of the mover **100**. The fourth and fifth pair of rollers **110** each engage one side of the first track portion **26** of the lower rail **20**. A sixth pair of rollers **110** are mounted on the side member **102** and extend above the second track portion **28** of the lower rail. The rollers **110** act together to engage the various surfaces of the rails **20** to both allow the mover **100** to travel along the rails **20** and to maintain the orientation of the mover **100** with respect to the track **10**. According to the illustrated embodiment, a position magnet **130** is mounted within the top member **104** and is configured to align with a sensor **150** mounted within the top surface **11** of the track segment.

(29) With reference to both FIGS. 5 and 7, a linear drive system is incorporated in part on each mover **100** and in part within each track segment **12**, **14** to control motion of each mover **100** along the segment. Coils **50** (see also FIG. 8) mounted along the length of the track **10** serve as first drive members. Each mover **100** includes a second drive member **140** which is configured to interact with electromagnetic fields generated by the coils **50** to propel the mover **100** along the track **10**. It is contemplated that the drive members **140** on each mover **100** may be drive magnets, steel back iron and teeth, conductors, or any other suitable member that will interact with the electromagnetic fields generated by the coils **50**. Commonly, the drive member **140** on each mover **100** includes permanent magnets which emit a magnetic field. The magnetic field generated by the drive member **140** on each mover **100** improves the mover interaction with the electromagnetic field generated by the coils **50** in comparison to a magnetically salient structure that has no magnetic field. For convenience, the invention will be discussed with respect to drive magnets **140** being used as the drive member within each mover **100**. However, it is understood that the other magnetically salient structures may be employed without deviating from the scope of the invention.

(30) With reference to FIG. 8, a series of coils **50** are positioned along the length of the track **10**. Each mover **100** includes at least one drive magnet **140** configured to interact with an electromagnetic field generated in the coils. Successive activation of the coils **50** establishes a



moving electromagnetic field that interacts with the magnetic field generated by each permanent magnet **140** mounted on the movers **100** and that causes the mover **100** to travel along the track **10**. Controlled voltages are applied to each coil **50** to achieve desired operation of the movers. The drive magnets **140** are mounted on the inner surface of the side member **102** and when mounted to the track **10** are spaced apart from a series of coils **50** extending along the track **10**. As shown in FIGS. 5 and 7, an air gap **141** is provided between each set of drive magnets **140** and the coils **50** along the track **10**. According to the illustrated embodiment, each coil **50** is placed in a channel **23** extending longitudinally along one surface of the track segment **12**. The electromagnetic field generated by each coil **50** spans the air gap **141** and interacts with the drive magnets **140** mounted to the mover **100** to control operation of the mover **100**.

(31) Turning next to FIG. 9, a portion of another exemplary independent cart transport system for moving articles or products is illustrated. The illustrated system includes a track **310** made up of multiple segments **312**. Rather than traveling along the sides of the track, as shown in FIG. 1, the illustrated movers **100** travel along a channel in the track **310**. The channel is defined by a bottom surface **316** and a pair of opposing side walls **313**. Rails **314** are placed along the length of the upper surface of each side wall **313** and are configured to support and engage the mover **100** as it travels along the track **310**. Power is delivered to segments **312** via a DC bus **320** extending along the track **310**. The DC bus **320** includes a positive rail **322** and a negative rail **324** where any suitable voltage potential is provided between the positive and negative rails to energize the coils **50**.

(32) The portion of the system illustrated in FIG. 9 includes two straight segments **312** and further illustrates an exemplary control system connected to the independent cart transport system. A segment controller **51** is provided within each track segment **312** to regulate current flow to the coils **50** forming the portion of the linear drive system in each track segment **312**. Optionally, each segment controller **51** may also be responsible for all, or a portion of, the motion control of each mover **100** as it travels along the corresponding segment **312**. According to one embodiment of the invention, the segment controllers **51** may be mounted together in a control cabinet. A cable, multiple cables, separate conductors, or a combination thereof extend from the control cabinet to each segment **12**, **14** to deliver current to the coils **50** and to receive feedback signals, for example, from position sensors **150**. In smaller systems, each segment controller **51** and an industrial controller **200** may be included in a single control cabinet. Depending on the size and layout of the track **10**, additional control cabinets may be distributed around the track and a portion of the segment controllers **51** located in a control cabinet proximate the track segment **12**, **14** they control. Separate control cabinets and controllers within a cabinet are communicatively connected via the network medium **160**. Although illustrated as blocks in FIG. 9 external to the track segments **312**, the illustration is to facilitate illustration of interconnects between controllers. According to still another embodiment, it is contemplated that each segment controller **51** may be mounted in the lower portion **319** of the track segment **312**. Each segment controller **51** is in communication with an adjacent segment controller **51** and a central controller **170** which is, in turn, in communication with an industrial controller **200**. According to yet another embodiment, the central controller **170** may be removed and the functions of the central controller **170** incorporated into the segment controllers **51**, the industrial controller **200**, or a combination thereof, and each segment controller **51** may communicate directly with the industrial controller **200**.

(33) The industrial controller **200** may be, for example, a programmable logic controller (PLC) configured to control elements of a process line stationed along the track **10**. The process line may be configured, for example, to fill and label boxes, bottles, or other containers loaded onto or held by the movers **100** as the travel along the line. In other embodiments, robotic assembly stations may perform various assembly and/or machining tasks on workpieces carried along by the movers **100**. The exemplary industrial controller **200** includes: a power supply **202** with a power cable **204** connected, for example, to a utility power supply; a communication module **206** connected by a

network medium **160** to the other controllers **51**, **170**; a processor module **208**; an input module **210** receiving input signals **211** from sensors or other devices along the process line; and an output module **212** transmitting control signals **213** to controlled devices, actuators, and the like along the process line. The processor module **208** may identify when a mover **100** is required at a particular location and may monitor sensors, such as proximity sensors, position switches, or the like to verify that the mover **100** is at a desired location. The processor module **208** transmits the desired locations of each mover **100** to a central controller **170** or to the respective segment controllers **51** where the receiving controller operates to generate commands for the current required in each coil **50** of the corresponding segment controller **51** to control motion of each mover **100**. Optionally, the industrial controller **200** may include a module in one of the slots of the chassis or embedded as a routine executing within the processor module **208** to perform a portion of the command generation and the processor module **208** may transmit a current command to a segment controller rather than a desired location.

(34) With reference also to FIG. **10**, each module in the industrial controller **200** may include its own memory and processor and be configured to execute one or more routines corresponding to the desired operation of the respective module. The portion of the industrial controller illustrated in FIG. **10**, shows a first processor **207** and a first memory device **209** located in the processor module **208** and a second processor **203** and a second memory **205** located in the communication module **206**. A backplane connects each module within the industrial controller **200** and backplane connectors **201a**, **201b** are shown connecting the two modules. Although illustrated as directly connecting the two modules, the backplane is a communication bus extending along the chassis of the industrial controller and each backplane connector **201** for a module engages a complementary backplane connector on the communication bus aligned with the slot on the chassis in which the module is inserted. A communication interface **199** within the communication module **206** is configured to connect to the industrial network **160**.

(35) The central controller **170** includes a processor **174** and a memory device **172**. It is contemplated that the processor **174** and memory device **172** may each be a single electronic device or formed from multiple devices. The processor may be a microprocessor. Optionally, the processor **174** and/or the memory device **172** may be integrated on a field programmable array (FPGA) or an application specific integrated circuit (ASIC). The memory device **172** may include volatile memory, non-volatile memory, or a combination thereof. An optional user interface **176** may be provided for an operator to configure the central controller **170** and to load or configure desired motion profiles for the movers **100** on the central controller **170**. Optionally, the configuration may be performed via a remote device connected via a network and a communication interface **178** to the central controller **170**. It is contemplated that the central controller **170** and user interface **176** may be a single device, such as a laptop, notebook, tablet or other mobile computing device. Optionally, the user interface **176** may include one or more separate devices such as a keyboard, mouse, display, touchscreen, interface port, removable storage medium or medium reader and the like for receiving information from and displaying information to a user. Optionally, the central controller **170** and user interface may be an industrial computer mounted within a control cabinet and configured to withstand harsh operating environments. It is contemplated that still other combinations of computing devices and peripherals as would be understood in the art may be utilized or incorporated into the central controller **170** and user interface **176** without deviating from the scope of the invention.

(36) The central controller **170** includes one or more programs stored in the memory device **172** for execution by the processor **174**. The central controller **170** can receive instructions for coordinating with industrial processes or machines. In one aspect, known as “centralized” control, the central controller **170** can determine one or more motion profiles for the movers **100** to follow along the track **10**. A program executing on the processor **174** is in communication with each segment controller **51** on each track segment via a network medium **160**. The central controller **170** may

transfer a command signal to the segment controller **51** in each track segment to control energization of the coils **50**. The central controller **170** may receive feedback signals corresponding to the identification and/or location of movers **100** along each track segment and control motion of the movers **100** directly from the central controller **170**. In one embodiment of the invention, it is contemplated that the central controller **170** may be implemented within the industrial controller **200** as either a portion of the control program executing in the processor module **208** or as a dedicated motion control module inserted within one of the slots of the industrial controller **200**.

(37) In another aspect, known as “distributed” control, the central controller **170** may be configured to transfer the desired motion commands, or a portion thereof, from the central controller **170** to each of the segment controllers **51**. The motion commands identify one or more desired movers **100** to be positioned at or moved along each track segment **312**. The central controller **170** may distribute motion commands to each segment controller **51** according to the mover **100** located at or proximate to the track segment **312**. The corresponding segment controller **51** for each track segment **312** may, in turn, determine the appropriate command signals for each mover **100** and transmit the command signals to one or more power segments in each track segment to control energization of the coils **50**. Distributed control can minimize the amount of communication in the system by allowing segment controllers **51**, rather than the central controller **170**, to control driving each mover **100** along the track **310**. In one embodiment of the invention, it is contemplated that the central controller **170** may be implemented within the industrial controller **200** as either a portion of the control program executing in the processor module **208** or as a dedicated motion control module inserted within one of the slots of the industrial controller **200**.

(38) A position feedback system provides knowledge of the location of each mover **100** along the length of the track segment **12**, **14** to the segment controller **51**. In one embodiment, the position feedback system can include one or more position magnets **130** mounted to the mover **100** and an array of sensors **150** spaced along the track segment **12**, **14**. With reference again to FIG. **1**, for convenience, only a few position sensors **150** are illustrated along one track segment **12**. It is contemplated that the position sensors **150** would continue along each track segment **12**, **14** and for the entire length of the track **10**. The sensors **150** are positioned such that each of the position magnets **130** are proximate to the sensor as the mover **100** passes each sensor **150**. The sensors **150** are a suitable magnetic field detector including, for example, a Hall Effect sensor, a magneto-diode, an anisotropic magnetoresistive (AMR) device, a giant magnetoresistive (GMR) device, a tunnel magnetoresistance (TMR) device, fluxgate sensor, or other microelectromechanical (MEMS) device configured to generate an electrical signal corresponding to the presence of a magnetic field. The magnetic field sensor **150** outputs a feedback signal provided to the segment controller **51** for the corresponding track segment **12** on which the sensor **150** is mounted. The feedback signal may be an analog signal provided to a feedback circuit **58** which, in turn, provides a signal to the processor **52** which corresponds to the magnet **130** passing the sensor **150**.

(39) The segment controller **51** also includes a communication interface **56** that receives communications from the central controller **170**, from adjacent segment controllers **51** in a path, and the industrial controller **200**. The communication interface **56** extracts data from the message packets on the communication network and passes the data to a processor **52** executing in the segment controller **51**. The processor may be a microprocessor. Optionally, the processor **52** and/or a memory device **54** within the segment controller **51** may be integrated on a field programmable array (FPGA) or an application specific integrated circuit (ASIC). It is contemplated that the processor **52** and memory device **54** may each be a single electronic device or formed from multiple devices. The memory device **54** may include volatile memory, non-volatile memory, or a combination thereof. The segment controller **51** receives the motion profile or desired motion of the movers **100** and utilizes the motion commands to control movers **100** along the track segment **312** controlled by that segment controller **51**.

(40) Each segment controller **51** generates switching signals to generate a desired current and/or

voltage at each coil **50** in the track segment **312** to achieve the desired motion of the movers **100**. The switching signals **72** control operation of switching devices **74** for the segment controller **51**. According to the illustrated embodiment, the segment controller **51** includes a dedicated gate driver module **70** which receives command signals from the processor **52**, such as a desired voltage and/or current to be generated in each coil **50** and generates switching signals **72**. Optionally, the processor **52** may incorporate the functions of the gate driver module **70** and directly generate the switching signals **72**. The switching signals **72** are provided to the power conversion segment in each track segment **312**, where each power conversion segment includes multiple power switching devices **74**. The switching devices **74** may be a solid-state device that is activated by the switching signal, including, but not limited to, transistors, thyristors, or silicon-controlled rectifiers.

(41) In one embodiment, the processor **52** can also receive feedback signals from sensors providing an indication of the current operating conditions within the power segment or of the current operating conditions of a coil **50** connected to the power segment. According to the illustrated embodiment, the power segment includes a voltage sensor **62** and a current sensor **60** at the input of the power segment. The voltage sensor **62** generates a voltage feedback signal and the current sensor **60** generates a current feedback signal, where each feedback signal corresponds to the operating conditions on the positive rail **322**. The segment controller **51** also receives feedback signals corresponding to the operation of coils **50** connected to the power segment. A voltage sensor **153** and a current sensor **151** are connected in series with the coils **50** at each output of the power section. The voltage sensor **153** generates a voltage feedback signal and the current sensor **151** generates a current feedback signal, where each feedback signal corresponds to the operating condition of the corresponding coil **50**. The processor **52** executes a program stored on the memory device **54** to regulate the current and/or voltage supplied to each coil and the processor **52** and/or gate driver **70** generate switching signals **72** which selectively enable/disable each of the switching devices **74** to achieve the desired current and/or voltage in each coil **50**. The energized coils **50** create an electromagnetic field that interacts with the drive magnets **140** on each mover **100** to control motion of the movers **100** along the track segment **12**.

(42) Turning next to FIG. **11**, multiple sensors **360** may be mounted to each mover **100** to monitor status of the mover during operation. According to the illustrated embodiment, each sensor **360** is mounted proximate to one of the rollers **110** for the mover **100**. The sensor **360** is configured to generate a feedback signal **365** corresponding to a force applied to the respective roller **110** by which it is mounted. The sensor **360** may be, for example, an accelerometer, a gyroscope, or a strain gauge. A force applied to the mover **100** via the linear drive system will generate an acceleration or deceleration of the mover measurable by the accelerometer. The forces experienced by mover **100** as it travels, for example, from the linear drive system, from friction along the rails, or from a load applied to the mover **100** may cause some rotational motion of the mover **100** measurable by the gyroscope and typically provided as a roll, pitch, and yaw of the mover. The force applied to the mover **100** via the linear drive system may cause some deflection (either expansion or contraction) of the material from which the mover **100** is constructed at the location at which the sensor **360** is mounted and which is measurable by the strain gauge. Thus, providing sensors at these locations generates a feedback signal corresponding to a force applied to the mover as seen at the location of the sensor.

(43) As also illustrated in FIG. **11**, it is contemplated that one or more coordinate systems may be defined for the independent cart system. A first coordinate system **420** may be a primary reference coordinate system, having a first x-axis, X, a first y-axis, Y, and a first z-axis, Z. This primary reference coordinate system may be used by all devices within the independent cart system or alternate coordinate systems may be utilized according to the application requirements. Each mover **100**, for example, may have a mover coordinate system **425** assigned to the mover. An origin for the mover coordinate system **425** is located at a desired position with respect to the mover, and a second x-axis, X.sub.m, a second y-axis, Y.sub.m, and a second z-axis, Z.sub.m, are defined with

respect to the origin of the mover coordinate system **425**. Further, each sensor **360** may have a sensor coordinate system **430** defined with respect to the location at which the sensor is located. An origin for the sensor coordinate system **430** is located where the sensor is mounted and, as illustrated, may be at the center of each sensor. A third x-axis, X.sub.s, a third y-axis Y.sub.s, and a third z-axis, Z.sub.s, are defined with respect to the origin of the sensor coordinate system **430**. The feedback signals **365** generated by each sensor may be assigned to one of the coordinate systems and translated between coordinate systems according to defined offsets between origins of the coordinate systems.

(44) As previously indicated, the sensor **360**, according to one aspect of the invention, may be a strain gauge **400**. With reference also to FIG. **12**, an exemplary circuit for the sensor **360** as a strain gauge **400** is illustrated. The illustrated embodiment is an electrical strain gauge constructed from a wheatstone bridge **405** having four resistors. One resistor is a variable resistor, where the resistance changes as a function of the strain present at the location. A conductive trace **404** applied to a foil material **402** is mounted over the location of interest. If strain causes a deformation in the location of interest, the foil **402** mounted on the location is either elongated or compressed as a result of the deformation. The elongation or compression of the foil **402** causes a similar change to the conductive trace **404** mounted on the foil **402**. Elongation and compression of the conductive trace **404** will alter the length and width of the conductive trace **404** thereby varying the resistance along the length of the trace **404**. A voltage applied across the variable resistance is provided to an amplifier **410**, which, in turn, generates the feedback signal **365**, corresponding to the strain present on the location. A varying resistance in the strain gauge **400** will cause a varying output voltage, V<sub>out</sub>, corresponding to the varying resistance, to the measured strain, and to force applied at the location.

(45) With reference again to FIG. **11**, each sensor **360** generates a feedback signal **365** provided to a control circuit **370** mounted on the mover **100**. An enclosure may be provided around the control circuit **370** to protect the control circuit from contamination in a manufacturing environment. With reference also to FIG. **13**, the control circuit **370** includes an interface **385** to receive each of the feedback signals **365**. It is contemplated that the interface **385** may include analog-to-digital (A/D) converters, amplifiers, buffers, multiplexers, and the like to convert voltage signals output from each sensor **360** to a digital value. Optionally, each sensor **360** may be configured to transmit data in a serial bit stream or as part of a data packet. The interface **385** may include a complementary communication interface to receive the digital data and provide the data values to a controller **375**.

(46) The controller **375** is configured to assemble the data received from each sensor into a data packet for transmission from the mover **100**. The controller **375** may be a microcontroller, a microprocessor, a field programmable gate array (FPGA), application specific integrated circuit (ASIC), or the like. It is contemplated that the controller **375** is configured to execute a series of instructions stored in memory, where the memory is either on board the controller or is an external device. The controller **375** may be configured to assemble the data directly into a data packet for transmission according to a desired communication protocol. Optionally, the controller **375** may be configured to perform some initial processing, for example, converting a measured strain to a force, as discussed below. The controller **375** may then assemble the processed data into a data packet for transmission according to the desired communication protocol.

(47) The data packet is provided to a transmitter **390** and antenna **395** mounted on the mover to transmit the data from the sensor **360** to a controller located remotely from the mover **100**. One or more receivers **350** (see also FIGS. **9-10**), may be positioned along the track **10**. In some applications, a single receiver **350** may have sufficient range and bandwidth to communicate with each mover **100** traveling along the track **10**. In other applications, the number of movers **100** may generate a volume of data packets that exceed the bandwidth of a single receiver **350** or the distance of travel along the track **10** may exceed the range of the receiver. As illustrated in FIG. **9**, each segment controller **51** may include or have a receiver **350** connected to the segment controller

**51.** The transmitter **390** transmits data packets containing either the measured value from the feedback signal or a processed data value, corresponding to the measured value, to a receiver **350** of the segment controller **51** in closest proximity to the mover **100**. The receiver **350**, in turn, transmits the data to the segment controller **51** or to a central controller **170** or industrial controller **200** in communication with the segment controller **51**.

(48) The control circuit **370** also includes a power circuit **380** mounted on the mover **100**. The power circuit **380** is configured to supply power to the sensor **360**, controller **375**, and transmitter **390**. According to one embodiment of the invention, the power circuit **380** may include a battery configured to supply power. One or more voltage regulators receive power from the battery and supply a regulated DC voltage, for example, at 3.3 VDC, 5 VDC, 12 VDC, 24 VDC, any other required DC or AC voltage, or a combination thereof. Optionally, the mover **100** includes a pickup device configured to travel along with the mover and to receive power transmitted from a power source external to the mover. The power source may be, for example, a supply coil or supply rail which receives a current that generates an electromagnetic field. The pickup device may be a coil mounted on the mover in which a secondary current is induced as the mover travels through the electromagnetic field. The secondary current is supplied to a voltage regulator circuit to provide the required voltages to each of the devices on the mover or to an energy storage device to store excess energy not required by the devices. If more power is required than may be provided from the pick-up device, energy stored in the energy storage device may supplement the energy supplied from the pick-up coil.

(49) The feedback signals **365** are transmitted to a controller located remote from the mover **100**. The controller may be the segment controller **51** controlling the section of track on which the mover is located. Optionally, the controller may be the central controller **170** for the independent cart system or the industrial controller **200** controlling operation of the track and external actuators. It is further contemplated that a dedicated receiver **350** or multiple receivers may be positioned around the track **10** to receive the feedback signals **365** from each mover **100** and the dedicated receiver **350** may relay the information to one of the controllers. The feedback signals may be utilized to monitor bearing wear and/or predict remaining life of bearings at each roller **110** on a mover.

(50) In operation, the force applied to the mover **100** and operating conditions of the mover are monitored at multiple locations by sensors **360** mounted at each location on the mover. Each sensor **360** mounted on the mover **100** generates one or more feedback signals **365** corresponding to a force applied to the mover **100**. It is contemplated that each sensor **360** may be a single axis sensor, generating a feedback signal **365** corresponding to one of the axes of the sensor coordinate system **430** or each sensor **360** may be a multiple axis sensor, generating feedback signals **365** corresponding to two or all three axes of the sensor coordinate system **430**.

(51) According to one embodiment of the invention, the sensor **360** is a strain gauge **400** configured to monitor deflection of the mover **100** at the location on which the strain gauge is mounted. The feedback signal **365** corresponding to the strain measured by the sensor may be used to determine the force experienced by the mover **100** at the location of the sensor. With equation 1, the measured strain may first be converted to a stress experienced by the mover **100**. Young's Modulus is a material property of the mover **100** and is a known value based on the construction of the mover. The value of Young's Modulus for each mover **100** may be stored in memory of one of the controllers and used to convert the measured strain to the value of stress.

(52)  $\sigma_n = E_n \epsilon_n$  (1)

where:  $\sigma_{\text{sub}.n}$  is the stress determined at sensor “n”,  $\epsilon_{\text{sub}.n}$  is the strain measured at sensor “n”, and E is Young's Modulus for the material at the location of sensor “n”.

(53) After determining the stress with equation 1, the stress may be converted to a force experienced by the mover at the sensor location by using equation 2. The value of stress previously

determined is multiplied by the sectional area of the mover at the location of the sensor to determine a force observed at the location of the sensor.

$$(54) \quad \sigma_n = \frac{F_n}{A} \quad (2)$$

where:  $\sigma_{\text{sub.n}}$  is the stress determined at sensor “n”,  $F_{\text{sub.n}}$  is the force observed at the location of sensor “n”, and A is the sectional area of the mover at the location of section “n”.

(55) Having determined a force present at each bearing, a controller may be used to monitor real-time performance of a mover **100** or to track performance of the mover over time. According to one aspect of the invention, it is contemplated that the controller monitoring performance may be the segment controller **51** monitoring performance of each mover **100** as the mover **100** travels along the segment controller **51**. The segment controller **51** monitors the forces in real-time and may generate a warning or fault message to be transmitted to either the central controller **170** or the industrial controller **200** if a force exceeds a predetermined threshold. Optionally, the segment controller **51** may store value of the forces observed for each mover in the memory **54** for the segment controller. An initial value may be stored during commissioning and changes in the value over time may be monitored. When a difference in the measured value of force (observed under consistent operating conditions) exceeds a predefined threshold, the segment controller **51** may generate a warning or fault message to be transmitted to either the central controller **170** or the industrial controller **200**.

(56) According to another aspect of the invention, the controller monitoring performance may be the central controller **170** or the industrial controller **200**. If the receivers **350** are connected to each segment controller **51**, the segment controllers **51** may be configured to retransmit data packets containing data from the sensors **360** to either the central controller **170** or the industrial controller **200**. Optionally, the receivers **350** may be connected directly to the central controller **170** or the industrial controller **200** and the corresponding controller may receive data from the sensors **360** directly from each mover **100**. The central controller **170** or industrial controller **200** may be configured to store data from each of the multiple movers **100** in the independent cart system. The controller may monitor the data in real-time or over a duration of time in a manner similar to that discussed above for each segment controller.

(57) The controller configured to monitor performance of the movers **100** in the independent cart system may detect a number of different operating conditions based on the feedback signals from the sensors **360**. According to a first aspect of the invention, the controller may be configured to determine a misalignment between adjacent track segments **12**. As each mover **100** transitions between a first track segment **12**, **14** and a second track segment **12**, **14**, where the second track segment is adjacent to the first track segment, there should be little or no change in operating performance between track segments. Ideally, each track segment is manufactured identical to one another and there is no change in operating performance as a mover transitions between track segments. However, variations do exist between adjacent track segments due to manufacturing tolerances, assembly tolerances, and the like. In some instances, adjacent track segments **12**, **14** may have some misalignment between the two segments. A mover **100** will experience a spike in strain and acceleration as it transitions between the misaligned segments. If the value of the spike in strain and/or acceleration exceeds a predefined threshold, the controller may generate a warning or fault message alerting a technician to the misalignment between adjacent track segments.

(58) According to another aspect of the invention, the controller may be configured to determine remaining life of bearings on each roller **110**. Each roller **110** may include a ball bearing, roller bearing, or other suitable bearing according to application requirements. The controller may be configured to monitor values detected by strain gauges **400** to determine forces applied to the bearings and further provide an indication of remaining bearing life. With reference to FIGS. **14-16**, orientation of the track **10** may impact bearing life. With reference also to FIGS. **6** and **11**, it may be observed that rollers **110** engage different portions **26**, **28** of the rails **20** with different

orientations. A combination of the orientation of each roller **110** on the mover **100** and the orientation of the track **10** will cause different rollers **110** to experience greater forces from gravity,  $g$ , than other rollers. With the track mounted in a vertical orientation, as shown in FIGS. **14** and **16**, the rollers **110** oriented horizontally in FIG. **6** will experience a greater force due to gravity,  $g$ , than the rollers **110** oriented vertically in the same figure. With the track mounted in a horizontal orientation, shown in FIG. **15**, the rollers **110** oriented vertically in FIG. **6** and positioned on the lower portion of the mover **100** will engage an upper surface of the portion **28** of the rail **20** protruding from the track segment and experience the greatest force due to gravity,  $g$ . Without the application of any other force, the bearings on rollers **110** experiencing greater forces due to gravity will wear quicker than the bearings on other rollers **110**. Measurements of the strain observed on the mover by each roller **110** provide an indication of the force experienced by each roller **110** and, in turn, by the bearings on each roller.

(59) The controller may monitor the status of the rollers **110** observing the greatest forces and determine the remaining life of each roller and/or the bearing on each roller. In addition to forces generated by gravity, forces will be exerted on the mover **100** due to acceleration and loading of the mover **100**. The memory of the controller may include a look up table corresponding to expected life of a bearing as a result of various forces applied. The controller may maintain a record of the operation of each mover **100** and the forces measured to determine a remaining life of each bearing. When the remaining life drops below a first threshold, a warning message may be generated to alert a technician preventive maintenance is required. If the remaining life drops below a second threshold, a second warning message or a fault message may be generated indicating an imminent failure of the bearing and roller **110** is possible.

(60) According to another aspect of the invention, the orientation of the track may be determined by monitoring the feedback signals **365** from each sensor **360**. When the mover is stationary, a force due to gravity will generate a reading from an accelerometer corresponding to a negative nine and eight-tenths meters per second square ( $-9.8 \text{ m/s}^2$ ) in the direction of gravity.

(61) If a multi-axis accelerometer is mounted on the mover **100**, the controller may determine orientation of the track **10** based on the axis along which the accelerometer is detecting the  $-9.8 \text{ m/s}^2$  reading.

(62) According to still another aspect of the invention, the feedback signals **365** from the sensors **360** may be utilized to determine the presence of a payload on the mover **100**. With reference again to FIG. **11**, the illustrated mover **100** includes an opening **101** on the upper surface of the mover **100**. It is contemplated that the opening **101** has a threaded interior surface, and a fixture (not shown) may be secured to the opening **101** with a bolt, screw, or other threaded fastener. The fixture may be a platform on which a load may be set. Optionally, the fixture may include a gripper, vacuum member, clamp, or other such gripping member to secure a load on the mover **100**. The weight of a payload mounted on the mover **100** will exert a downward force on the top member **104** of the mover and a torque on the bolts **105** securing the top member **104** to the side member **102**. The force may cause deflection on the top member **104** and/or side member **102** and, in particular near the edges of each member where the top and side members are joined. A strain gauge mounted proximate each roller **110** along the top of the side member **102** or at the rollers **110** along the inward side of the top member **104** will detect the deflection due to the presence of a load on the mover **100**. Further, a greater load will cause a greater deflection. The controller (segment controller **51**, central controller **170**, or industrial controller **200**) configured to receive the feedback signal **365** from the strain gauge **400** may utilize a look-up table or execute a calculation to determine a weight of the payload as a function of the additional strain measured when loaded. Thus, each strain gauge may be used not only to detect the presence of a load on each mover, but further be utilized to determine the amount of loading present on the mover **100**.

(63) It is yet another aspect of the invention, the feedback signals **365** may be utilized to monitor wear in the independent cart system. During commissioning or during early operation of the mover



**100**, a first value of the feedback signal **365** may be stored during a particular operation. For example, as the mover **100** travels across a particular location, the strain, stress, or acceleration may be monitored and stored in memory. Optionally, multiple values of the feedback signal **365** may be stored as the mover **100** travels along a length of the track. During subsequent operation of the mover **100**, measured values of strain, stress, or acceleration, as observed in the feedback signals **365** may be compared to the stored values. A change in the feedback value over time that exceeds a predefined threshold may indicate excessive wear or damage that has occurred to the mover **100** or to the track segment **12, 14** over which the mover **100** is travelling. A warning or fault message may be generated to alert a technician to the change in value and to the mover **100** and/or to the track segment **12, 14** on which the change in value in occurred, providing an indication of the wear or damage in the independent cart system.

(64) Still another aspect of the invention allows a controller to determine improvements to a motion profile being executed by the mover **100**. As discussed above, each mover **100** is controlled by the linear drive system to travel along the track. The industrial controller **200**, central controller **170**, segment controller **51**, or a combination thereof is operational to generate a motion profile for each mover. The segment controller **51** controls operation of the coils **50** spaced along the track to achieve the desired motion of each mover **100**. As a mover **100** travels along the track **10**, the feedback signals **365** from each sensor **360** may be periodically stored to obtain a corresponding measured acceleration profile, velocity profile, stress profile, or strain profile corresponding to the motion profile generated by the controller. One of the controllers analyzes the feedback signals **365** stored in memory to identify the greatest value of the acceleration, velocity, stress, or strain. This maximum value of the measured feedback may be compared to a first predefined threshold.

Optionally, the controller may analyze the feedback signals **365** to identify a duration of time along the profile during which the values exceed a second predefined threshold. When the measured performance of the mover **100** exceeds an instantaneous maximum value or when the measured performance of the mover **100** exceeds a second maximum value for a predefined duration, the controller may generate a message alerting a technician to the performance. Optionally, the controller may be further configured to adapt the motion profile being generated to reduce the maximum value or to reduce the sustained value over the duration, thereby reducing the overall stress or strain on the system.

(65) It should be understood that the invention is not limited in its application to the details of construction and arrangements of the components set forth herein. The invention is capable of other embodiments and of being practiced or carried out in various ways. Variations and modifications of the foregoing are within the scope of the present invention. It also being understood that the invention disclosed and defined herein extends to all alternative combinations of two or more of the individual features mentioned or evident from the text and/or drawings. All of these different combinations constitute various alternative aspects of the present invention. The embodiments described herein explain the best modes known for practicing the invention and will enable others skilled in the art to utilize the invention.

(66) In the preceding specification, various embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

## Claims

1. A method for monitoring bearings of a mover in an independent cart system, comprising the steps of: generating a feedback signal from at least one sensor mounted on the mover, wherein the at least one sensor is mounted proximate a bearing on the mover and wherein the feedback signal

corresponds to a force present at the bearing; transmitting the feedback signal from the at least one sensor to a controller for the independent cart system; comparing the feedback signal from the at least one sensor to a predetermined threshold; and generating a message when the feedback signal from the at least one sensor exceeds the predetermined threshold.

2. The method of claim 1, wherein the message is a fault message corresponding to a failure of the bearing proximate to the at least one sensor.

3. The method of claim 1, further comprising the steps of: storing an initial value of the feedback signal; determining a difference between the initial value of the feedback signal and the feedback signal; and generating a message corresponding to a remaining life of the bearing as a function of the difference between the initial value of the feedback signal and the feedback signal.

4. The method of claim 1, further comprising the steps of: receiving the feedback signal from the at least one sensor at a control circuit mounted on the mover; generating a data packet including a value for the feedback signal from the at least one sensor; and the step of transmitting the feedback signal from the at least one sensor to the controller includes transmitting the data packet from the control circuit to the controller.

5. The method of claim 1, wherein the independent cart system includes a track having a plurality of segments, the method further comprising the step of receiving a position feedback signal at the controller, the position feedback signal corresponding to a location of the mover along the track.

6. The method of claim 5, further comprising the step of identifying a misalignment between adjacent track segments at the controller as a function of the position feedback signal and the feedback signal from the at least one sensor.

7. The method of claim 5, further comprising the step of identifying an orientation of the track in the controller as a function of the feedback signal from the at least one sensor.

8. The method of claim 1, further comprising the steps of: storing an initial value of the feedback signal corresponding to an unloaded state of the mover; detecting a load present on the mover when a difference between a present value and the initial value of the feedback signal exceeds a second predetermined threshold.

9. The method of claim 1, further comprising the step of determining a weight of a load present on the mover as a function of the feedback signal from one of the plurality of sensors.

10. The method of claim 1, wherein the controller is operative to control operation of the mover along a track for the independent cart system, the method further comprising the steps of: comparing the feedback signal from the at least one sensor to a second predetermined threshold; and adjusting either an acceleration or a velocity of the mover to reduce a magnitude of the force present at the bearing when the feedback signal exceeds the second predetermined threshold.

11. A system for monitoring bearings of a mover in an independent cart system, the system comprising: at least one sensor mounted on the mover proximate a bearing on the mover, wherein the at least one sensor is configured to generate at least one feedback signal corresponding to a force present at the bearing; a transmitter mounted on the mover, wherein the transmitter is configured to transmit the at least one feedback signal to a controller for the independent cart system; the controller in communication with the transmitter to receive the at least one feedback signal, wherein the controller is operative to: compare the at least one feedback signal from the at least one sensor to a predetermined threshold; and generate a message when the at least one feedback signal from the at least one sensor exceeds the predetermined threshold.

12. The system of claim 11, wherein the message is a fault message corresponding to a failure of the bearing proximate to the at least one sensor.

13. The system of claim 11, wherein the controller further includes a memory and wherein the controller is further operative to: store an initial value of the at least one feedback signal in the memory; determine a difference between the initial value and a present value of the at least one feedback signal; and generate a message corresponding to a remaining life of the bearing as a function of the difference between the initial value and the present value of the at least one

feedback signal.

14. The system of claim 11, further comprising a control circuit mounted on the mover, wherein the control circuit is configured to: receive the at least one feedback signal from the at least one sensor; generate a data packet including a value of the at least one feedback signal; and output the data packet to the transmitter.

15. The system of claim 11, wherein: the independent cart system includes a track having a plurality of segments, the controller is further operative to receive a position feedback signal at the controller, the position feedback signal corresponding to a location of the mover along the track.

16. The system of claim 15, wherein the controller is further operative to identify a misalignment between adjacent track segments at the controller as a function of the position feedback signal and the at least one feedback signal from the at least one sensor.

17. The system of claim 15, wherein the controller is further operative to identify an orientation of the track in the controller as a function of the at least one feedback signal from the at least one sensor.

18. The system of claim 11, wherein the controller further includes a memory and wherein the controller is further operative to: store an initial value of the at least one feedback signal corresponding to an unloaded state of the mover; and detect a load present on the mover when a difference between a present value and the initial value of the at least one feedback signal exceeds a second predetermined threshold.

19. The system of claim 11, wherein the controller is further operative to determine a weight of a load present on the mover as a function of the at least one feedback signal from one of the plurality of sensors.

20. The system of claim 11, wherein the controller is further operative to: control operation of the mover along a track for the independent cart system, compare the feedback signal from the at least one sensor to a second predetermined threshold, and adjust either an acceleration or a velocity of the mover to reduce a magnitude of the force present at the bearing when the feedback signal exceeds the second predetermined threshold.

---